

A FINITE ELEMENT CALCULATION OF STRESS INTENSITY FACTORS BY A MODIFIED CRACK CLOSURE INTEGRAL

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Abstract—An efficient technique for evaluating stress intensity factors is presented. The method, based on the crack closure integral, can be used with a constant strain finite element stress analysis and a coarse grid. The technique also permits evaluation of both Mode I and Mode II stress intensity factors from the results of a single analysis. Example computations are performed for a double cantilever beam test specimen, a finite width strip with a central crack, and a pin loaded circular hole with radial cracks. Close agreement between numerical results given by this approach and reference solutions were found in all cases.

INTRODUCTION

THE FINITE element method has become a useful technique for solving linear elastic fracture mechanics problems. Many approaches have been described in the literature. As examples, surveys on the topic have been conducted by Gallagher[1], by Jerram and Hellen[2] and by Oglesby and Lomacky[3]. Benzley and Parks[4] have reviewed some of the computer programs currently being used.

The finite-element approaches to fracture mechanics can be described as either direct or as indirect methods. In the direct method, the stress intensity factors are calculated directly from the solution. In the indirect method, an energy-release rate is calculated and stress intensity factors inferred from it. Direct methods can be used with either conventional (nonsingular) elements or with elements containing a stress singularity at the crack tip. In the former case, it is necessary to curve fit the stress or displacement behavior away from the crack tip and extrapolate to the crack tip to determine the stress intensity factor. In the latter case, it is calculated directly. For example, see Ref. [7]. Shapes of elements include triangular[8], quadrilateral[9] and polygonal[10].

The indirect approach determines the stress intensity factor by evaluating the energy-release rate. Several techniques for evaluating energy-release rates have been described in the literature. These methods are based on computing the change in potential energy for two crack lengths[6, 11] or computing a change in compliance for two crack lengths[12, 13]. Other approaches are based on evaluating the change in the stiffness matrix due to crack extension and require only one analysis[14-16]. The J integral evaluated for a linear-elastic material, is identical to the energy-release rate and offers a convenient way of evaluating $\mathcal{G}$ with a single analysis[17].

Although the energy release rate, $\mathcal{G}$, is defined in terms of the behavior at the crack tip, $\mathcal{G}$ can be evaluated by the following equation for cases where the energy is dissipated only at the crack tip.

$$\mathcal{G} = \frac{dW}{dc} - \frac{dU}{dc} \quad (1)$$

In eqn (1), U is the strain energy of the body per unit thickness, W is the work of the external tractions per unit thickness, and c is the crack length. The relationship between $\mathcal{G}$ and the stress intensity factor for an isotropic material in plane strain is established using Irwin's crack closure integral[18]. This is

$$\mathcal{G} = \frac{(1-\nu^2)}{E} K_I^2 + \frac{(1-\nu^2)}{E} K_{II}^2 \quad (2)$$

Thus, generally, the indirect techniques do not give the Mode I and Mode II values for K

separately unless it is known *a priori* that the loading produces only a Mode I or Mode II response.

The method described here is based directly on computing the crack closure integral [18]. While the crack-closure integral serves as a very valuable philosophical tool in fracture mechanics—i.e. as the basis of eqn (2)—it appears to have received relatively little attention in the literature as a computational device. In one attempt to do so, Jerram reported results for Mode I conditions using the crack closure integral [19]. Two analyses were used in this technique. The first was for the structure with the prescribed loading conditions. In the second analysis, a unit force was placed at the node on the crack face nearest to the crack tip. The force required to close the crack was determined and the work to close the crack evaluated. Jerram's results were obtained with constant strain elements. Hellen and Blackburn [16] report that attempts to use elements with mid-side nodes (higher order elements) were not as successful and gave errors up to 20% when only the corner node was fixed.

If Jerram's suggested procedure were followed for general loading with Mode I and Mode II crack opening modes, it appears that three analyses would be required to evaluate K_I and K_{II} . In the modified crack closure integral presented here, only one analysis is required to obtain values for both K_I and K_{II} . This is accomplished by approximating the force to close the crack from the results of a single analysis.

This method presented here falls into the category of indirect methods. Unlike other indirect methods, this method permits both K_I and K_{II} to be evaluated from a single analysis. The method does not require that stresses be calculated. Conventional constant strain elements have been used successfully and a coarse grid near the crack tip was found to be sufficient. The following contains a description of the method and the results for selected problems. Results are compared with reference solutions taken from the literature.

DESCRIPTION OF THE METHOD

Irwin's contention is that if a crack extends by a small amount Δc , the energy absorbed in the process is equal to the work required to close the crack to its original length [18]. Using a polar coordinate system with the origin at the extended crack tip, this statement in equation form is

$$\mathcal{G} = \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \int_0^{\Delta c} \sigma_y(\Delta c - r, 0) \bar{v}(r, \pi) dr + \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \int_0^{\Delta c} \tau_{xy}(\Delta c - r, 0) \bar{u}(r, \pi) dr \quad (3)$$

where $\mathcal{G}$ is the energy-release rate, σ_y and τ_{xy} are the stresses near the crack tip, $\bar{u}$ and $\bar{v}$ are the relative sliding and opening displacements between points on the crack faces and Δc is the crack extension at the crack tip.

The first and second integrals of eqn (3) are expressions for the energy release rates for Mode I and Mode II deformations, respectively. Thus, eqn (3) can be expressed as the following two equations.

$$\mathcal{G}_I = \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \int_0^{\Delta c} \sigma_y(\Delta c - r, 0) \bar{v}(r, \pi) dr \quad (4)$$

and

$$\mathcal{G}_{II} = \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \int_0^{\Delta c} \tau_{xy}(\Delta c - r, 0) \bar{u}(r, \pi) dr \quad (5)$$

where $\mathcal{G}_I$ and $\mathcal{G}_{II}$ are the energy release rates for Modes I and II.

Expressions relating $\mathcal{G}_I$ to K_I are obtained by substituting Westergaard's representations for the stresses and displacements in the vicinity of a crack into eqns (4) and (5) and taking the limit. The results are

$$\mathcal{G}_I = \frac{K_I^2}{E} \beta \quad (6)$$

and

$$\mathcal{G}_{II} = \frac{K_{II}^2}{E} \beta \quad (7)$$

where $\beta = 1$ for plane stress and $1 - \nu^2$ for plane strain and E is Young's modulus.

The finite element computer program AXISOL was selected for the analysis of the configurations considered here. In this analysis, the displacements have a linear variation over the elements, and the stiffness matrix is written in terms of forces and displacements at the element nodes. To be consistent with this finite element representation, the approach for evaluating $\mathcal{G}_I$ and $\mathcal{G}_{II}$ [from eqns (4) and (5)] is based on the nodal forces and displacements. An illustration of the method is described in terms of Fig. 1. The physical interpretation of the integrals in eqns (4) and (5) is the amount of work required to close the crack and amount, Δc . In terms of the finite element representation, this is one-half the product of the forces at c and d required to close these nodes. In equation form, the expressions for $\mathcal{G}_I$ and $\mathcal{G}_{II}$ are

$$\mathcal{G}_I = \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \bar{F}_c \cdot (v_c - v_d) \quad (8)$$

and

$$\mathcal{G}_{II} = \lim_{\Delta c \rightarrow 0} \frac{1}{2\Delta c} \bar{T}_c \cdot (u_c - u_d). \quad (9)$$

For computational purposes, a value of Δc is selected as shown in Fig. 1. The value of $\bar{F}_c$ and $\bar{T}_c$ are taken to be the y - and x -forces, respectively, that are required to hold nodes c and d together. When evaluating $\bar{F}_c$ and $\bar{T}_c$, it is convenient to maintain l_2 equal to Δc and Elements (1), (2), (3) and (4) the same size. This is not necessary, however, and a modification to handle this case is described in the appendix. For the problems considered here, the forces at the crack tip (F_e and T_e in Fig. 1) were obtained by placing very stiff springs between points e and f and evaluating the forces in these springs. For cases where Δc was not equal to l_2 , estimates of $\bar{F}_c$ and $\bar{T}_c$ were obtained from eqns (IV) and (V) of the Appendix.

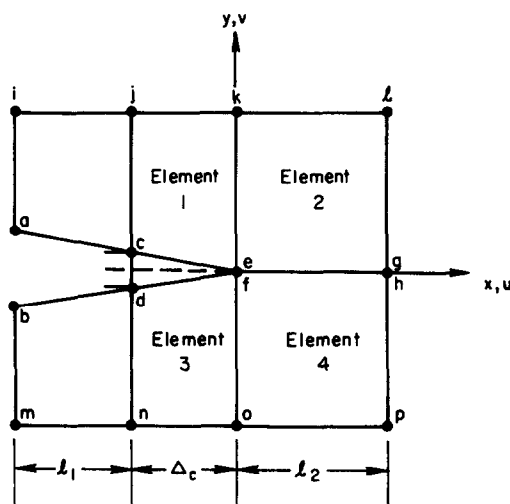


Fig. 1. Finite-element nodes near crack tip.

NUMERICAL RESULTS

The method was applied to three different problems: the double cantilever beam test specimen, a finite width strip containing a central crack, and a bolt fastened double lap joint containing radial cracks. The double cantilever beam (DCB) test specimen is shown in Fig. 2. The loading is a parabolic shear stress distribution at the end of the beam. Three ratios of c/h

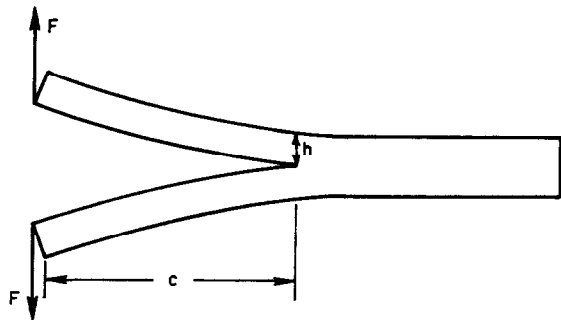


Fig. 2. Double cantilever beam test specimen.

were considered. Stress intensity factors for these cases were obtained by the method described here and by three other techniques. The results are given in Table 1. The results shown in Column 1 were obtained by solving two problems; one with crack length c and the other with crack length $c + \Delta c$. The energy-release rate was evaluated from the difference in potential energies for the two problems. Stress intensity factors were obtained from the relationship between $\mathcal{G}$ and K_I in eqn (6). The second column contains values obtained from the approximate solution given by Kanninen[20]. Values in the third column were obtained by evaluating the energy release rate from the J -integral expression described in Ref. [17]. The fourth column gives the results obtained by modified crack closure integral method described here.

The second configuration considered was a finite width strip containing a central crack. Figure 3 shows the dimensions of the plate. A finite element grid for one quadrant of the plate is shown in Fig. 4. A value of $\Delta c = 0.1$ was used here. Calculations were performed for six ratios of c/w . Two remote loading conditions were considered; uniform σ_x for Mode I and uniform τ_{xy} for Mode II. Results for Mode I and Mode II are given in Table 2 along with reference values. Reference results given in Columns 1 and 3 were obtained by the same techniques used to obtain Columns 1 and 3 for the double cantilever beam test specimen configuration. The results

Table 1. Mode I stress intensity factors for double cantilever beam shown in Fig. 2 ($\Delta c = 0.05$)

c/h	Energy Release Rate = $\Delta E/\Delta c$	Beam Model [20]	J Integral	Modified Crack Closure Integral
2.00	9.009	9.147	9.053	9.062
2.50	10.684	10.877	10.761	10.744
3.00	12.358	12.616	12.429	12.429

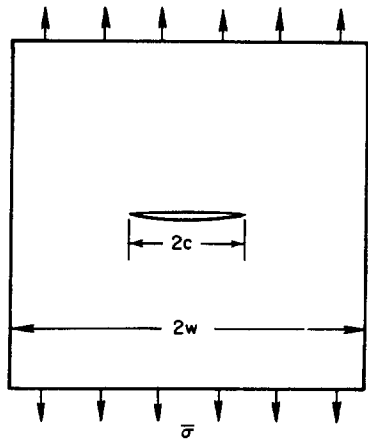


Fig. 3. Central crack configuration.

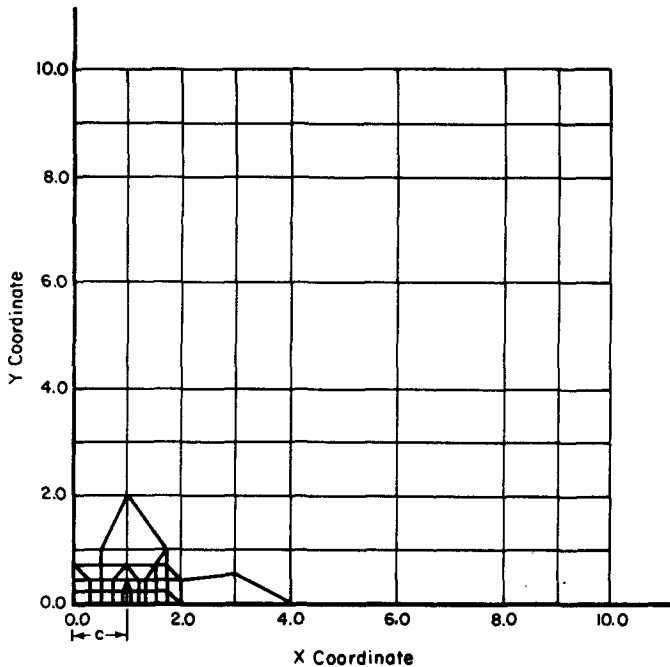


Fig. 4. Finite-element grid for one quarter of a plate containing an imbedded crack.

Table 2. Stress intensity factors for imbedded crack shown in Fig. 3 ($\Delta c = 0.1$)

c/w	Energy Release Rate = $\Delta E/\Delta c$	Reference [21]	J Integral	Modified Crack Closure Integral
<u>Mode I Values</u>				
.10	{ 1.743	1.783	1.706	1.707
.11		1.796	1.796	1.804
.30	{ 3.188	3.262	3.136	3.150
.31		3.320	3.208	3.217
.50	{ 4.637	4.713	4.584	4.584
.51		4.798	4.650	4.667
<u>Mode II Values</u>				
.10	--	1.772	1.714	1.707

in Column 2 were obtained from the equation given by Feddersen[21]. Results obtained by the modified crack closure integral described here are presented in Column 4.

The third configuration was a model of a bolt fastened double lap joint described in Ref. [22]. The finite element grid for this model is shown in Fig. 5. A two-dimensional finite representation was used for the wing panel and a separate finite element representation was used for the cover plates. The two grids were coupled through the fastener. Two radial cracks were modeled in the wing panel. Numerical results for the Mode I stress-intensity fastors, as a function of crack length, were obtained using the modified crack closure integral described here. Table 3 gives the results for six values of crack length ranging from 0.1 in. to 0.55 in. Reference values shown in Table 3 are from Ref. [22] and were obtained by curve fitting the crack opening displacement of the finite element nodes to Westergarrd's expressions for displacements near a crack tip in an isotropic material. The values of Δc used in the computations

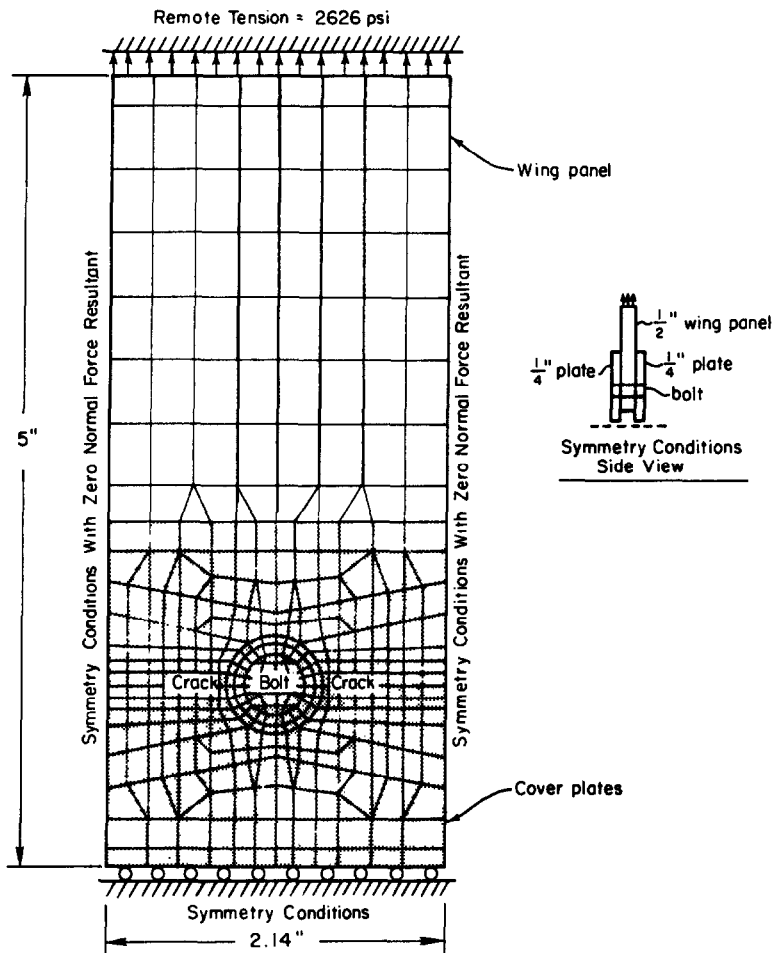


Fig. 5. Finite-element grid for a bolt fastened double lap joint from Ref. [22].

Table 3. Comparison of calculated and reference values for stress-intensity factors configuration in Fig. 5

Crack Length, C (inches)	Δc (inches)	Modified Crack Closure Integral K_I (ksi)	Reference (22) K_I (ksi)	$\frac{\Delta K_I}{K_I \text{ Ref.}}$
.10	.045	8.21	8.06	1.9%
.16	.05	8.43	8.72	3.3%
.26	.10	9.33	9.76	4.4%
.36	.10	9.91	10.26	3.4%
.46	.10	10.84	11.09	2.3%
.55	.09	11.54	11.95	3.4%

range from 0.045 in. to 0.1 in. For cases where Δc was not equal to l_2 , as defined in Fig. 1, an estimate of the force to close the two nodes immediately ahead of the crack tip was obtained from eqn (IV) of the Appendix.

A final study with the method was to examine the sensitivity of the solutions to the assumed incremental crack extension, Δc . The problem considered was a finite plate with a central crack. Figure 6 shows the ratio of the value of K_I obtained by the modified crack closure method described here to the reference value plotted against the ratio $\Delta c/c$. These results show that accuracies within 6% of the reference solution were obtained for values of Δc up to 20% of the crack length. Reference values for K_I were obtained from [21].

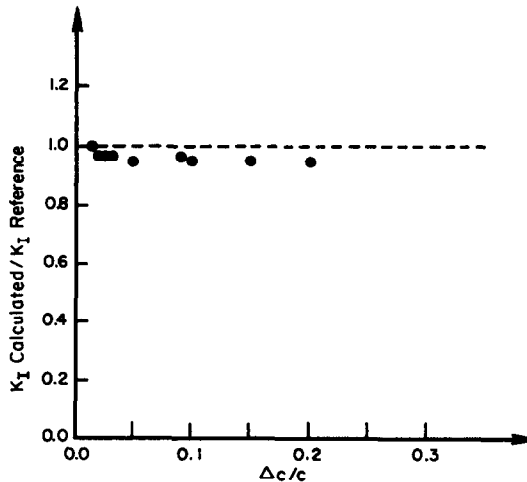


Fig. 6. Effect of Δc on the stress-intensity factor calculated by the modified crack-closure integral.

SUMMARY AND CONCLUSIONS

Stress intensity factors were evaluated using a modified crack closure integral and a constant strain finite element stress analysis program AXISOL. In this approach, K_I and K_{II} are determined from a single analysis by a small amount of computational effort. The method was applied to three configurations; a double cantilever beam test specimen, a finite width plate containing an imbedded crack, and a bolt fastened double lap joint containing two radial cracks. In all cases, numerical results for stress-intensity factors were compared with results obtained by independent techniques.

The results of three independent reference methods were presented for the double cantilever beam test specimen and the imbedded crack configurations. Closest agreement was between the method presented here and the J integral results. Both Mode I and Mode II calculations were done for the imbedded crack configuration. For the bolt fastened double lap joint configuration, reference solutions for one other method were presented. Values for K_I were calculated for six crack lengths and compared with reference values. The calculated values for K_I were within 4.5% of the reference values.

It is emphasized that this method was done with constant strain elements using nodal forces and nodal displacements and noted that poor results were obtained when the method was used with higher order elements. This is in agreement with the findings reported in Ref. [16]. For the problems considered, this approach provided results that were very close to the reference values. Hence, the method appears to offer a useful and straightforward technique for evaluating stress-intensity factors.

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APPENDIX

ESTIMATE OF FORCES TO CLOSE THE CRACK AT AMOUNT Δc

Let the forces F_e and T_e denote the forces required to hold nodes e and f together. Referring to Fig. 1, if $\Delta c = l_2$ and Elements (1), (2), (3) and (4) are the same size, and Δc is small compared to the crack length c , then it is reasonable to assume that a good estimate for the values of $\bar{F}_e$ and $\bar{T}_e$ will be F_e and T_e , respectively. However, if l_2 is not equal to Δc , then F_e and T_e represent the forces to close a crack by the amount l_2 . The approach taken here was to estimate $\bar{F}_e$ and $\bar{T}_e$ in the following way.

For isotropic materials, the stresses near the crack tip vary as $1/r^{1/2}$. Thus, the force over a length Δc is given by

$$F(\Delta c) = A_1 \int_0^{\Delta c} \frac{dr}{r^{1/2}} = 2A_1 \Delta c^{1/2}. \quad (\text{I})$$

Similarly, the force for a length l_2 is

$$F(l_2) = 2A_1 l_2^{1/2}. \quad (\text{II})$$

Taking the ratio of eqns (I) and (II) and solving for $F(\Delta c)$, the force for a length Δc is

$$F(\Delta c) = \left(\frac{\Delta c}{l_2} \right)^{1/2} F(l_2). \quad (\text{III})$$

In the calculations, the following estimates for $\bar{F}_c$ and $\bar{T}_c$ were used,

$$\bar{F}_c = \left(\frac{\Delta c}{l_2} \right)^{1/2} F_c \quad (\text{IV})$$

and

$$\bar{T}_c = \left(\frac{\Delta c}{l_2} \right)^{1/2} T_c. \quad (\text{V})$$

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